

2026년도 1학기 박사학위심사 요약자료

Development and Validation of a 3D-Printed SILPS for Local Skin Dosimetry

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- 04 Development and Experimental Validation of the SILPS**
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1.1 Background

- In Korea, dosimetry anomalies among radiation workers have steadily occurred.
- Although most cases were due to lost or damaged dosimeters, excessive exposure from radiation irradiating devices has also continued to occur.
- In such events, the skin is the first organ exposed and the tissue where deterministic effects directly appear.
- Several recent cases involved skin overexposure caused by X-ray irradiators.
- ICRP defines local skin dose as the dose averaged over the most highly exposed 1 cm² area at a depth of about 70 μm.
- However, conventional retrospective skin dosimetry methods do not account for the evaluation depth, ensure tissue equivalence, or reproduce the exposure posture, and still require post exposure readout.

1. Introduction

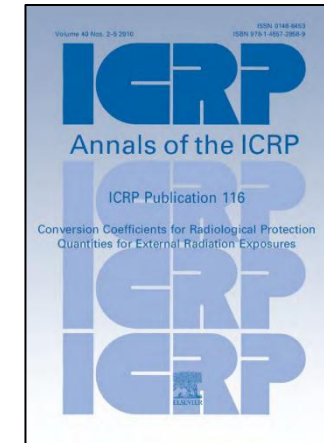
1.2 State of the art

1.2.1 ICRP's discussion of Local Skin Doses

1.2.2 Retrospective dosimetry

1.2.3 3D-printed plastic scintillator for dosimetry

- Recent discussions by the ICRP on local skin dose were reviewed.
- Conventional retrospective dosimetry methods and their limitations were reviewed.
- In addition, previous applications of 3D-printed plastic scintillators in dosimetry were reviewed.



< Existing Retrospective Dosimetry Method and Limitations >

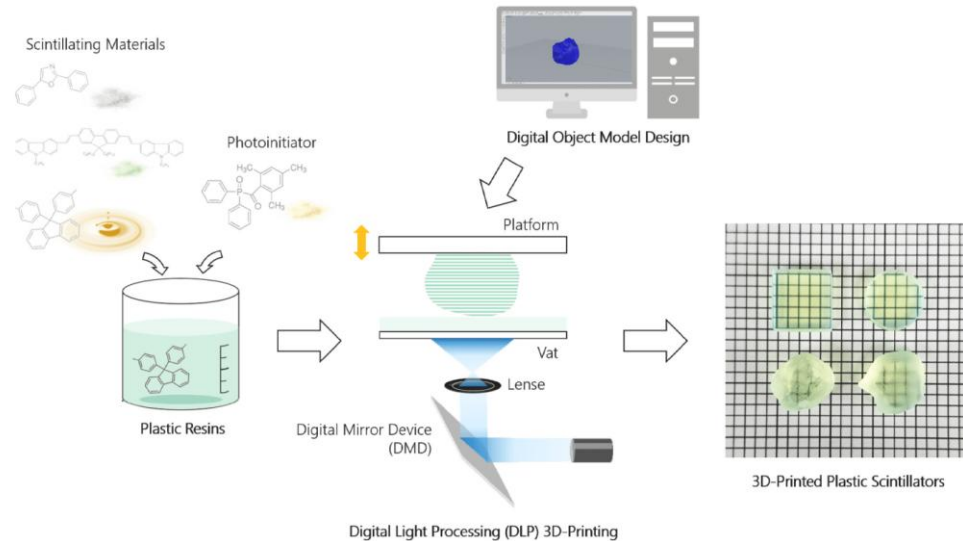
Retrospective Dosimetry methods		Limitations of Retrospective dosimetry methods
Computing simulations	Montecarlo simulations	○ Impossible to fully reconstruct the accident exposure conditions ○ Results depend on coding proficiency and require long calculation time
Biological Dosimetry	Chromosome Analysis/Blood Assay	○ Not applicable to local dose assessment ○ Time-consuming from cell culture to dose evaluation ○ Possible dose underestimation after long post-exposure intervals
Physical Dosimetry	Electron Paramagnetic Resonance(EPR)	○ Requires sample collection, causing additional injury and analysis time
	TL/OSL	○ Requires post-exposure readout ○ Sensitive to temperature, humidity, and visible light
Clinical Dosimetry	Doctor's diagnosis	○ High uncertainty in evaluated dose ○ Results depend on examiner's experience ○ Only applicable after deterministic effects become visible

2. 3D-Printed Plastic Scintillators and SIL Fabrication

2.1 Development of 3D printed plastic scintillator

2.1.1 Previous works and fabrication optimization

- Since the early 2020s, the development of 3D-printed Plastic Scintillators(PS) has attracted increasing attention.
- DLP-based vat photopolymerization allows rapid and economic fabrication of free-form plastic scintillators.
- The KOMERATE D0241 monomer-based PS is applied to this study because it has the best light yield and ease of handling in 3D printing methods.
- However, previous research revealed the combination of resin, but printing parameter discussions on complex shapes were not made.
- Therefore, printing parameters for precise PS production were optimized in this study.



Printing geometries	Optimized printing parameters					
	Base number	Base exposure time(s)	Body exposure time(s)	Slice thickness (μm)	Resin temperature(°C)	UV-intensity (mW/cm ²)
Cube(1×1×1cm ²)	3	25	17	100	35	550
Cylinder(Ø60, h:1 cm)	3	30	17	100	35	550
3D letters	3	40	17	100	35	550
Sphere(Ø25.4) with support interface(Ø1)	3	60	15	50	35	550
Thin film(50 μm)	1	25	-	50	35	550
Thin film(50 μm)	2	25	-	25	35	550

< DLP 3D printing-based plastic scintillator fabrication procedure >

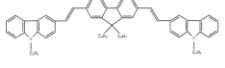
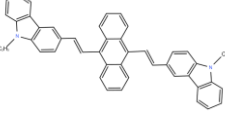
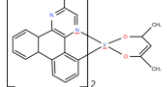
< Optimizing the Print Parameters of the RMPS470 Scintillator >

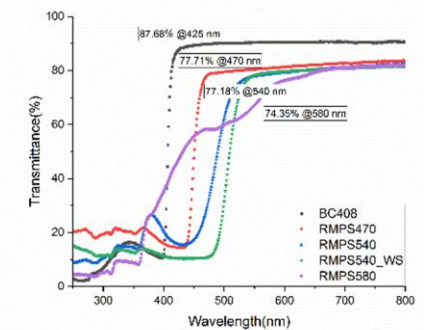
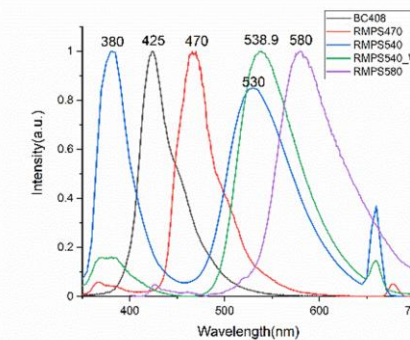
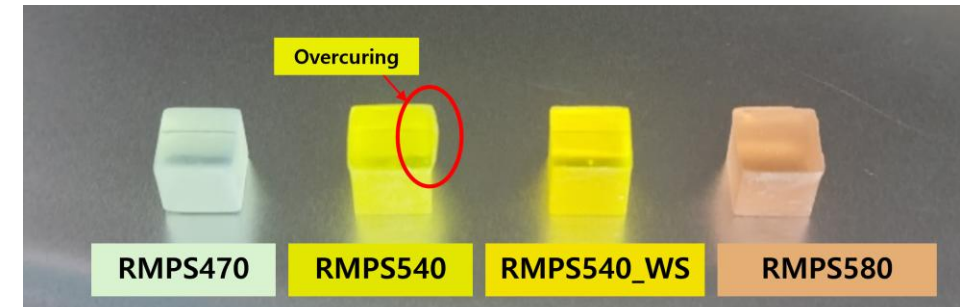
2. 3D-Printed Plastic Scintillators and SIL Fabrication

2.1 Development of 3D printed plastic scintillator

2.1.2 Long emission wavelength plastic scintillator

- RMPS470, with an emission peak at 470 nm, was optimized for high light yield.
- However, it is not well matched to photodetectors with peak sensitivity above 500 nm, such as silicon photomultipliers.
- Therefore, plastic scintillators with emission peaks at 540 nm and 580 nm(RMPS540, RMPS540_WS, and RMPS580)were developed and their properties were evaluated.

Plastic scintillator	Components				
	Monomer	Primary dye	Photo-initiator	Secondary fluor	
RMPS470	D0241 (Ethoxylated bisphenol fluorene diacrylate (OPPEA 40%))	PPO (Diphenyloxazole)	TPO (Diphenyl phosphine oxide)	ADS086BE (0.0075 wt%)	
RMPS540				ADS081BE (0.0075 wt%)	
RMPS540_WS				ADS081BE (0.0375 wt%)	
RMPS580				ADS076RE (0.0075 wt%)	



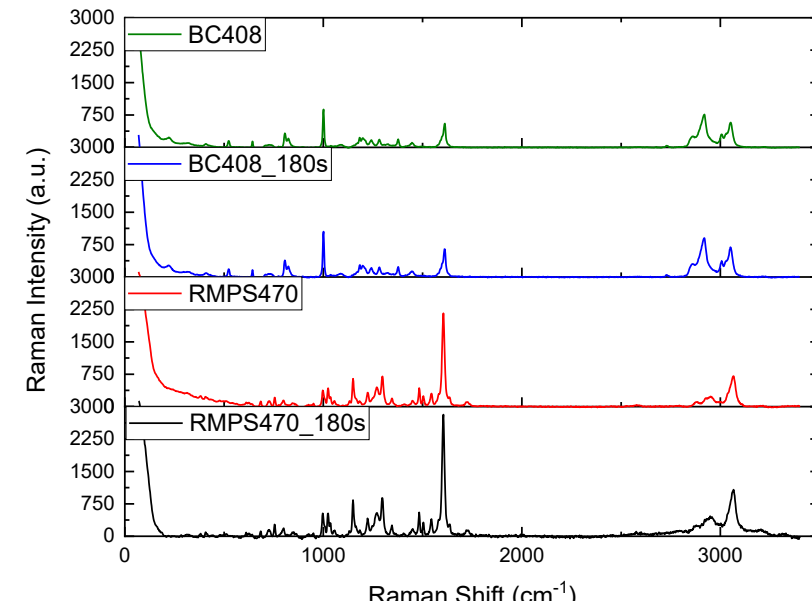
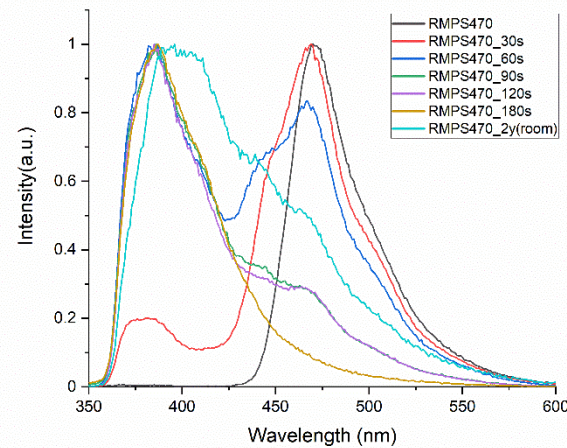
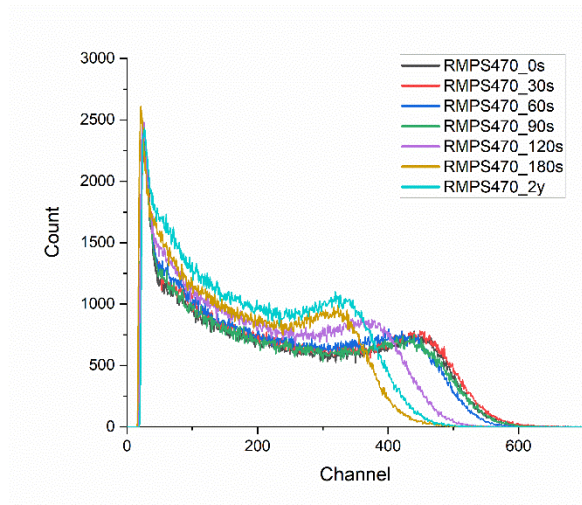
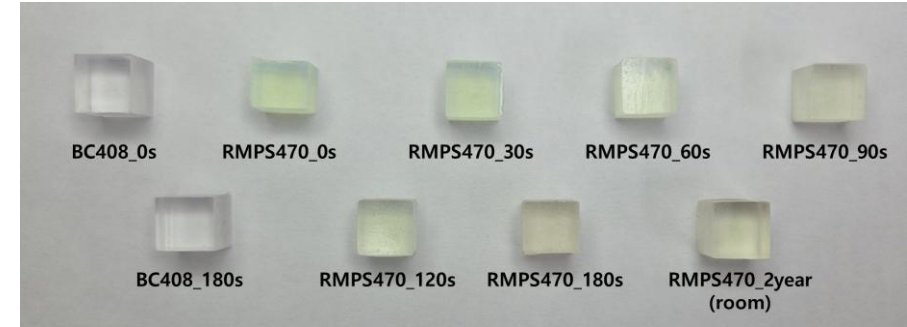
< Composition & characteristics of long emission wavelength plastic scintillator >

2. 3D-Printed Plastic Scintillators and SIL Fabrication

2.2 Stabilities of 3D printed plastic scintillator

2.2.1 Light-induced Bleaching effect

- Photobleaching can occur in some plastic scintillators, causing performance degradation under UV exposure.
- All evaluated 3D-printed plastic scintillators showed changes in emission wavelength and light yield.
- Therefore, all plastic scintillators require UV protection during storage.



< Photo-bleaching effect of plastic scintillators >

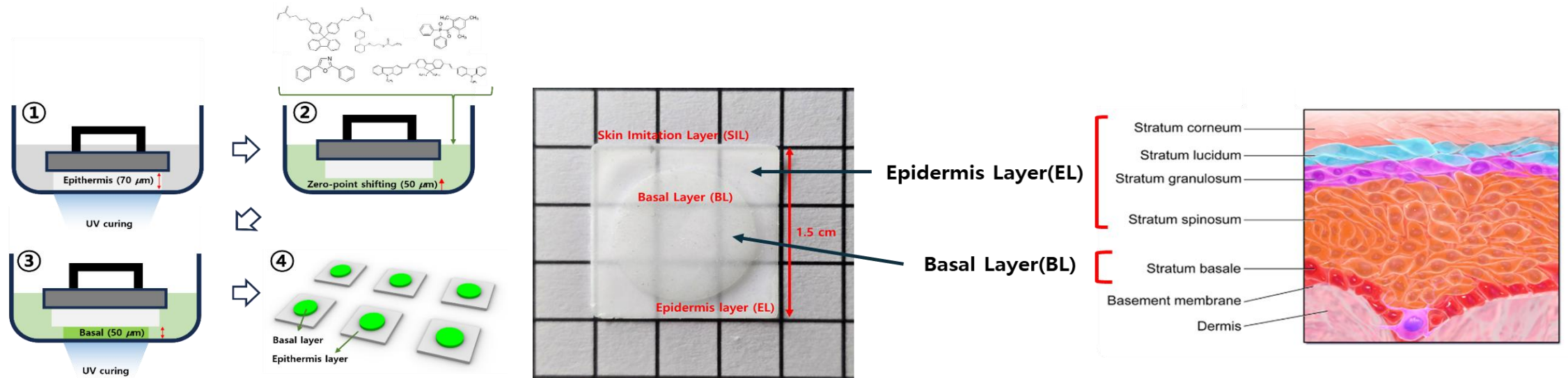
3. SIL Characterization and Tissue-Equivalence Assessment

3.1 Design and Fabrication of the Skin-Imitating Layer (SIL)

3.1.1 Structural Design of the SIL

3.1.2 Fabrication Procedure of the SIL

- The ICRP 89 skin model divides the skin into seven layers, from the stratum corneum to the dermis.
- The stratum basale is the proliferative cell layer where deterministic radiation effects in the skin occurs.
- The ICRP recommends evaluating superficial skin dose as the energy absorbed in a 50 μm thick, 1 cm^2 volume at an average depth of 70 μm .
- The 3D-printed Skin Imitating Layer (SIL) is a cell for local superficial skin dose assessment, in which the basal layer is made of plastic scintillator.



< Fabrication procedure of SIL >

< Structure of the skin >

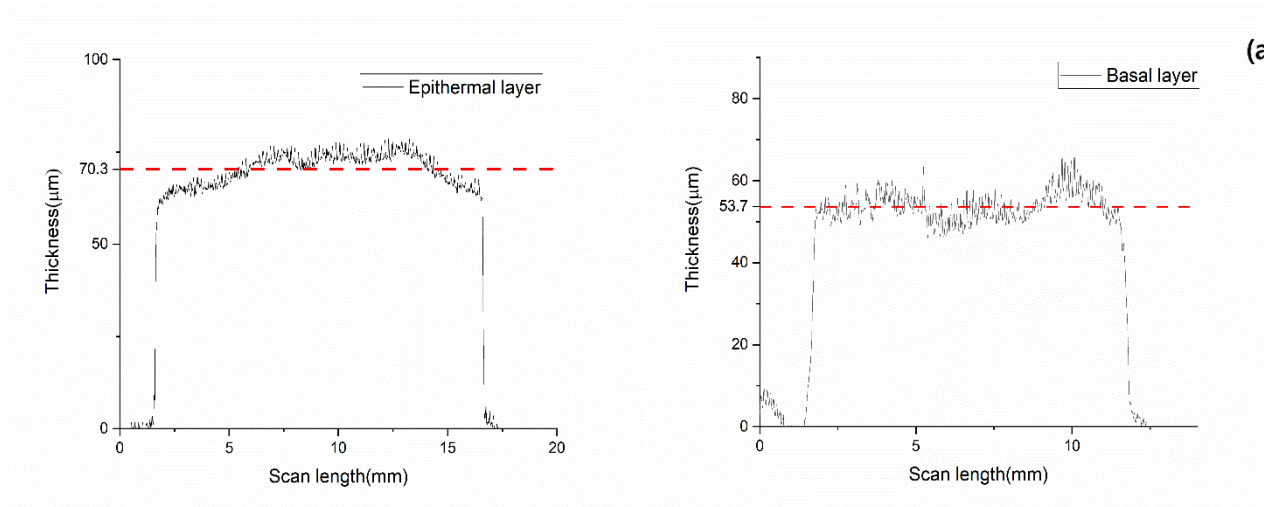
3. SIL Characterization and Tissue-Equivalence Assessment

3.2 Structural Characterization and Stability of the SIL

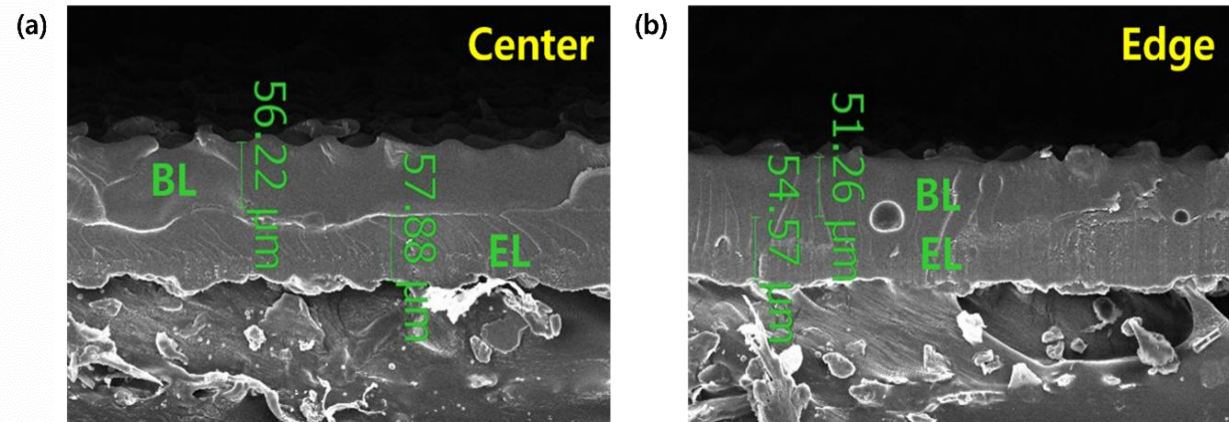
3.2.1 Thickness Control and Measurement

3.2.2 SIL degradation

- The target thickness of 50 μm in the 3D-printed SIL was verified using a surface profiler and SEM.
- The surface profiler provided the average thickness and thickness distribution from step-height measurements.
- SEM evaluated the EL-BL interface and the cross-sectional thickness at the center and edge.
- Both methods confirmed that the EL and BL were fabricated to 50 μm within the error range.



< Thickness distribution of SIL (surface profiler) >



< Thickness of EL & BL (SEM) >

3. SIL Characterization and Tissue-Equivalence Assessment

3.3 Tissue-Equivalence Assessment of the SIL

3.3.1 Elemental Composition Analysis of the SIL

3.3.2 Analytical Evaluation of Tissue Equivalence

- Accurate assessment of skin equivalent dose requires tissue equivalence, that is, a composition and density similar to those of human skin.
- The elemental compositions of the EL and BL were analyzed, and their relative density, electron density, and effective atomic number were compared with water and the ICRP 110 reference phantom material.

< Composition & Tissue equivalence of EL & BL >

Material		BL (RMPS470)	EL (CUKM25W Matt white)	Water	ICRP110 phantom material (skin)
Composition (wt%)	H	5.84	8.58	11.19	10.0
	C	76.85	58.20	-	19.9
	O	15.55	26.20	88.81	65.0
	N	0.34	4.12	-	4.2
	S	-	-	-	0.2
	Others	1.42	2.90	-	0.7
C/O		4.94	2.22	-	0.3
Relative density (g/cm3)		1.21	1.08	1	1.09
Electron density (10 ²³ electrons g ⁻¹)		3.14	3.18	3.35	3.29
Effective Z _r		6.18	6.37	7.41	7.13
Effective Z _k		5.74	5.72	6.59	6.33

$$N_e(\text{electrons/g}) = \sum_{n=C,H,O,N,S\cdots} \frac{N_A Z_n f_n}{A_n}$$

$$a_n = \frac{N_{e,n}}{N_e}$$

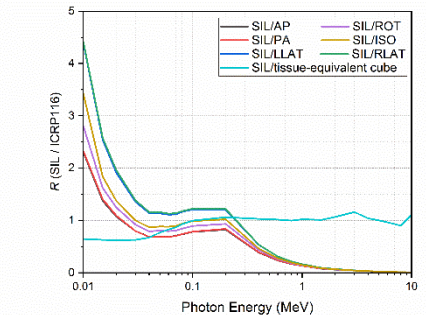
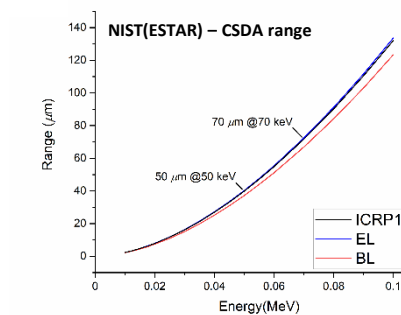
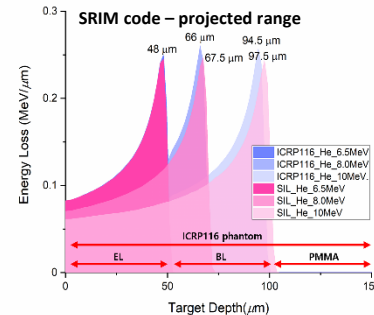
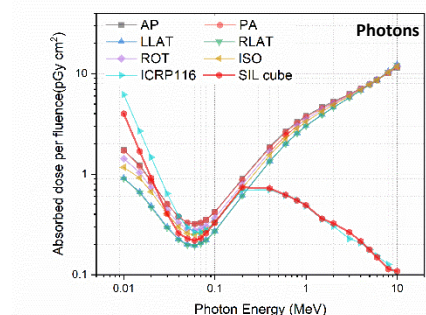
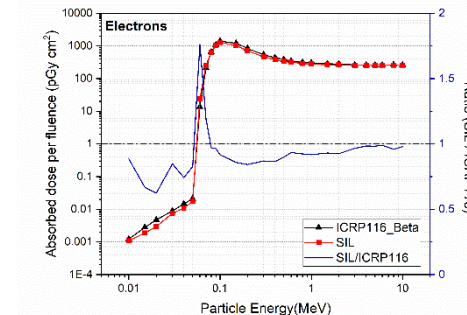
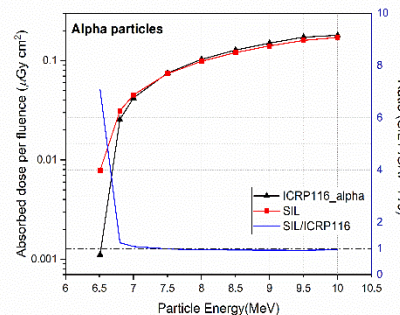
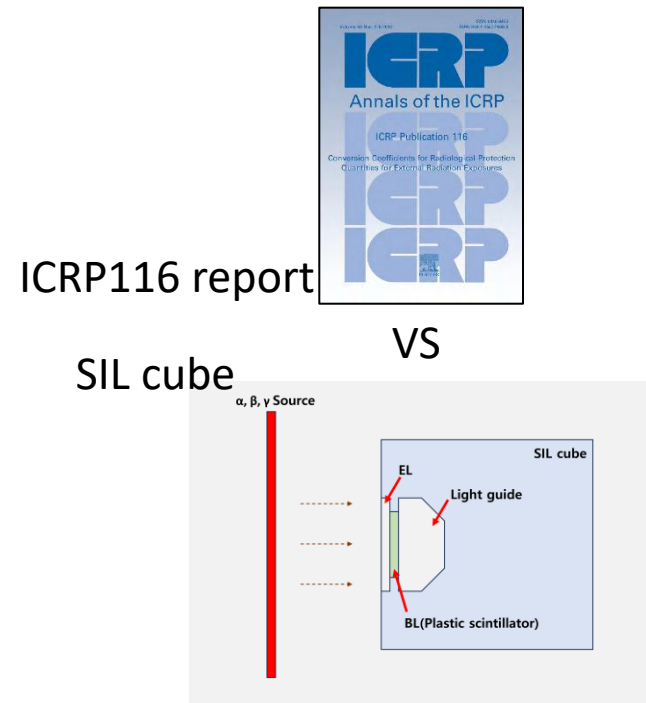
$$Z_{eff} = (a_1 Z_1^m + a_2 Z_2^m + a_3 Z_3^m + \cdots + a_n Z_n^m)^{1/m}$$

3. SIL Characterization and Tissue-Equivalence Assessment

3.3 Tissue-Equivalence Assessment of the SIL

3.3.3 Monte Carlo–Based Dosimetric Evaluation

- Monte Carlo simulations were also performed to evaluate the tissue equivalence of the SIL.
- ICRP 116 used MCNPX to calculate local skin dose per fluence (D/Φ) for alpha particles and electrons.
- Based on this model, benchmark calculations were carried out by applying the SIL to the ICRP 116 tissue-equivalent cube.
- For photons, since ICRP 116 provides only the average whole-skin dose for six irradiation geometries, the SIL results were compared with those of the tissue-equivalent cube.



< Monte Carlo–Based Tissue Equivalence Evaluation >

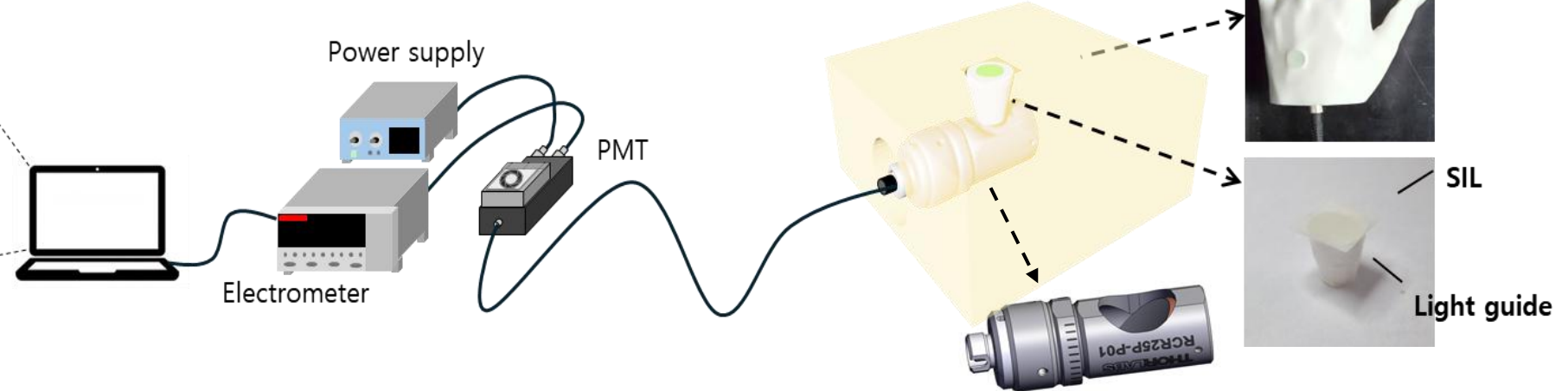
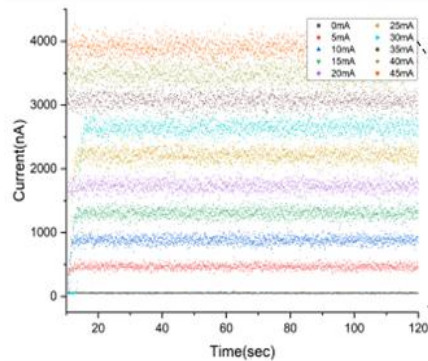
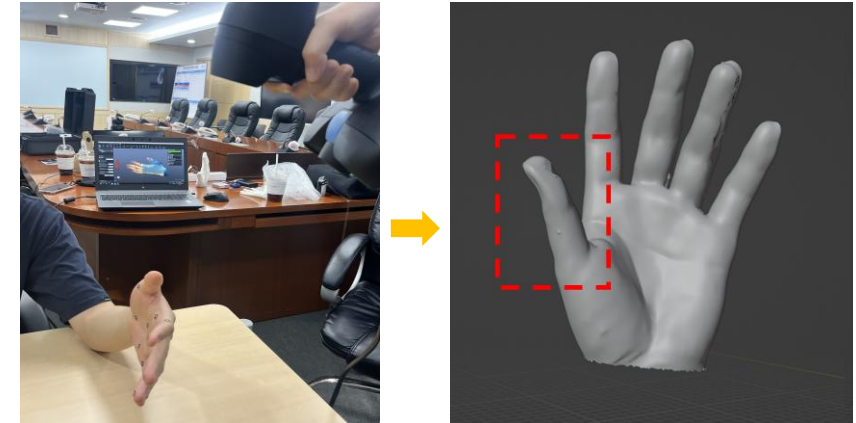
4. Development and Experimental Validation of the SILPS

4.1 System Architecture of the SILPS

4.1.1 System configuration

4.1.2 Anthropomorphic Implementation Strategy

- The local skin dose measurement system based on the SIL was named SILPS (Skin Imitating Layer Phantom System).
- SILPS consists of the SIL, light guide, reflective collimator, tissue-equivalent phantom, optical fiber, PMT, power supply, electrometer, and software.
- The tissue-equivalent phantom reproduces the exposed body region and posture using 3D scanning and 3D printing.
- The software converts the measured current into dose in real time.



< Schematic view of SILPS >

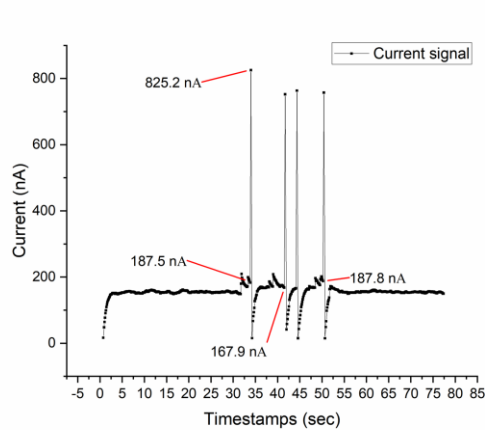
4. Development and Experimental Validation of the SILPS

4.2 Dosimetric Methods for the SILPS

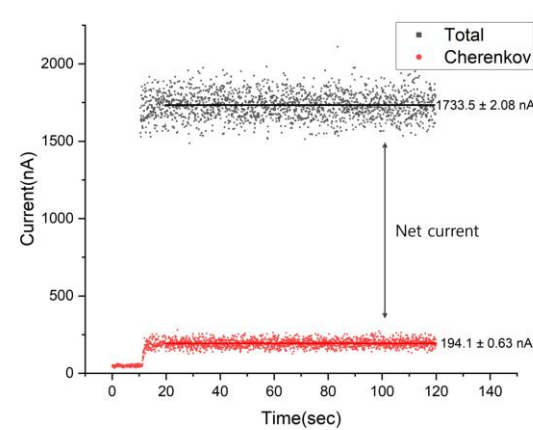
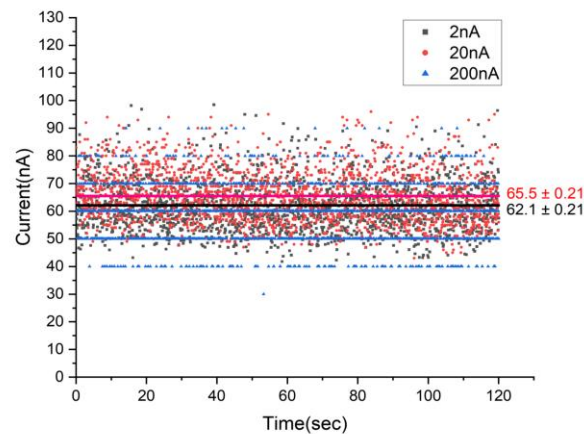
4.2.1 Current-Mode Readout

4.2.2 Separation of radioluminescence

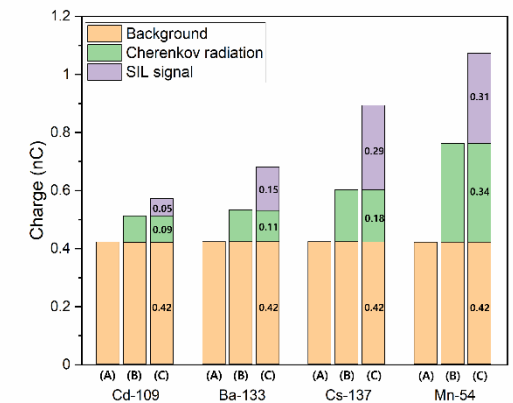
- The SILPS electrometer continuously measures the small current corresponding to skin dose rate.
- The current range and sampling interval were defined through preliminary experiments, and moving-average filtering was applied for noise reduction.
- To isolate the radioluminescence current generated in the light guide and optical fiber, two measurements were performed with and without the SIL.
- Additional experiments with four gamma-ray sources confirmed the radiation response of the BL and the clear separation of the radioluminescence signal.



< Current mode readout of SILPS >



< Cherenkov signal separation >



4. Development and Experimental Validation of the SILPS

4.3 X-ray based SILPS Calibration

4.3.1 X-ray irradiation Conditions

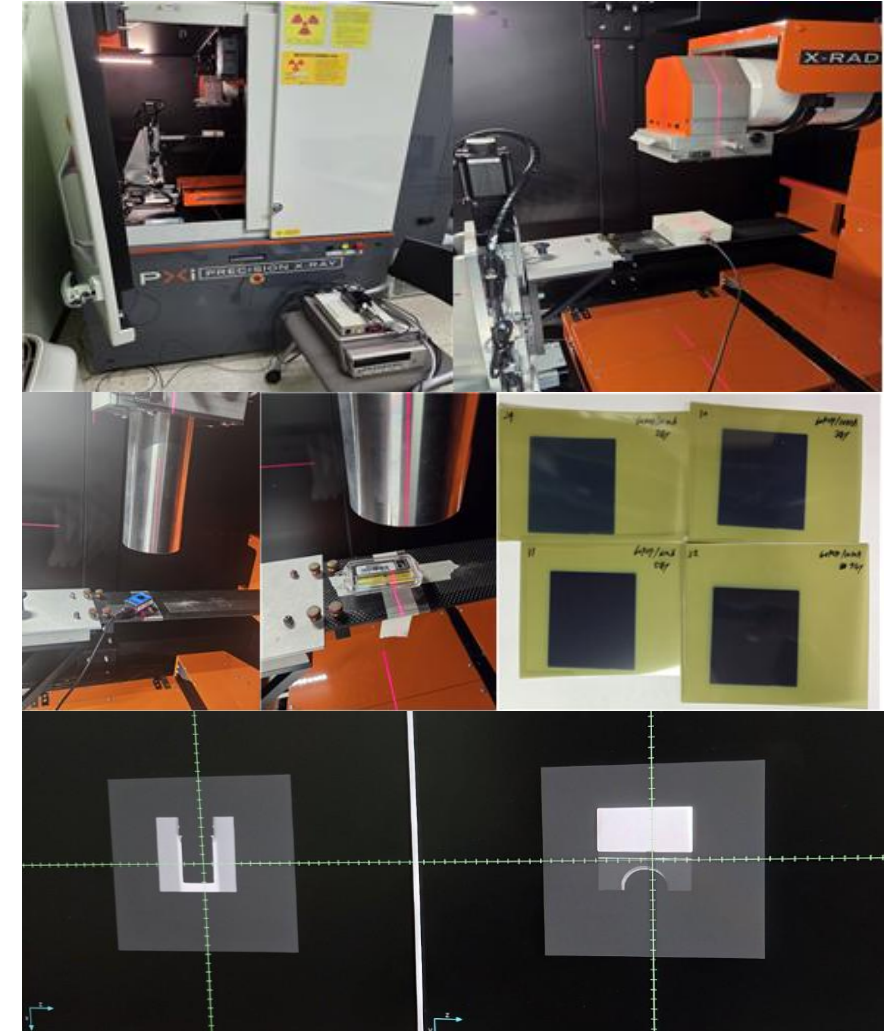
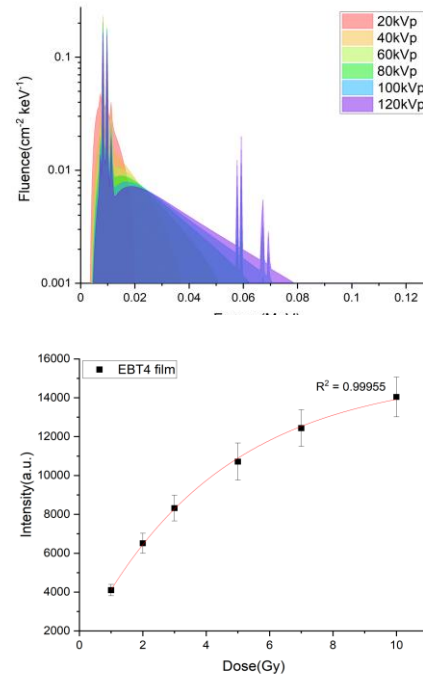
4.3.2 Reference Dosimetry Methods (Comparators)

- SILPS was calibrated for X-rays using an industrial X-ray irradiator over its main operating range(20-120 kVp).
- Under the same conditions, four conventional skin dose assessment methods* were also applied to determine dose and dose rate.

* Ionization chamber,
Gafchromic EBT4 film,
Thermoluminescent dosimeter(TLD),
MCNP6 code

Tube voltage (kVp)	Tube current (mA)
20, 40, 60	5, 10, 20, 30, 40
80	5, 10, 20, 30, 35
100	5, 10, 20, 30
120	5, 10, 20, 25

< Setup of X-ray irradiator >



< X-ray exposure to 3 methods & SILPS >

4. Development and Experimental Validation of the SILPS

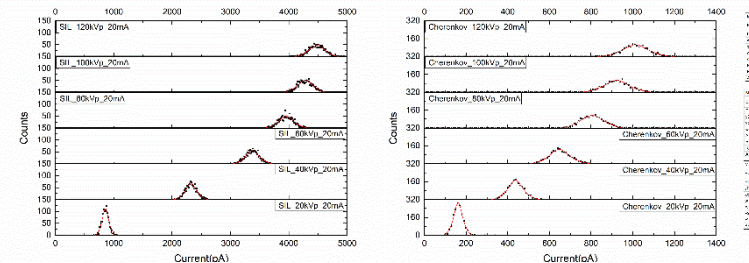
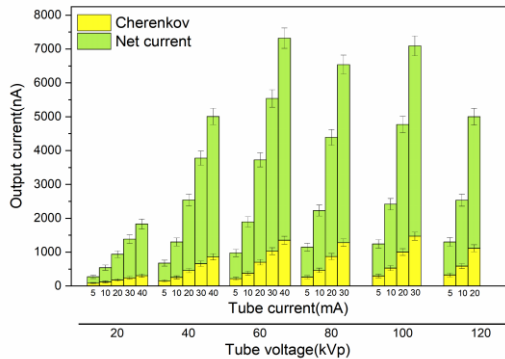
4.3 X-ray based SILPS Calibration

4.3.3 Dose Linearity

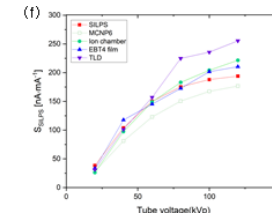
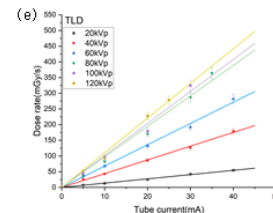
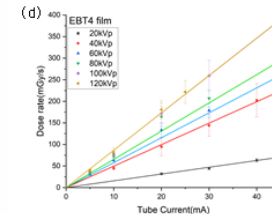
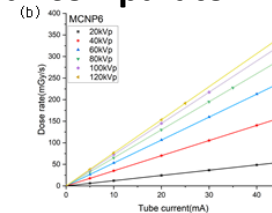
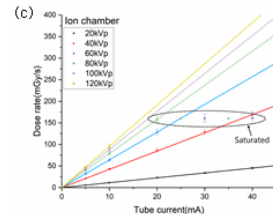
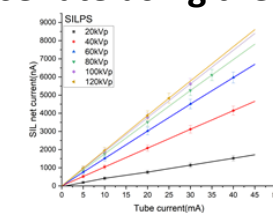
4.3.4 Comparison of Dosimetry Methods

4.3.5 Conversion from Current to Dose Rate

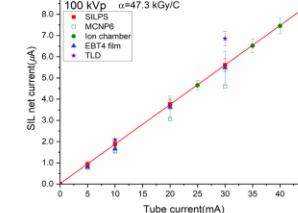
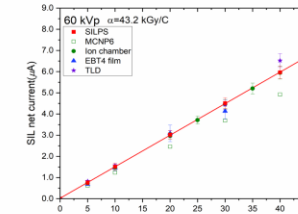
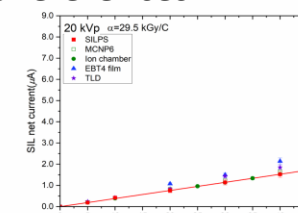
- SILPS showed the best linearity to tube current among the four comparator methods ($R^2 > 0.9999$).
- It enabled stable skin dose rate measurements up to 280 mGy/s with uncertainties below 10%.
- Its output current can also be calibrated to dose rate using the four comparator methods as references.



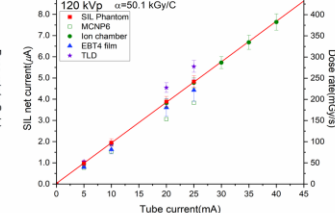
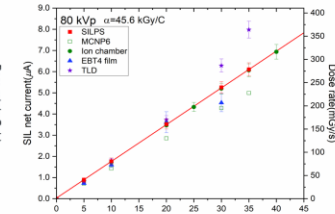
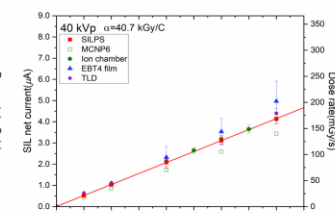
< SILPS net signal >



< Dose Linearity >



< SILPS calibration >



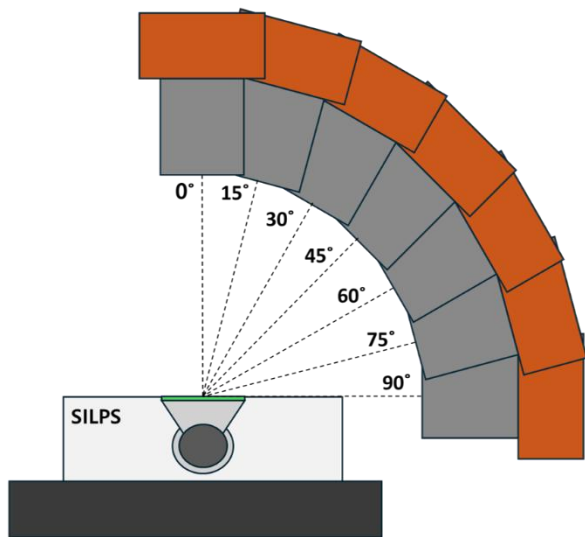
< SILPS calibration >

4. Development and Experimental Validation of the SILPS

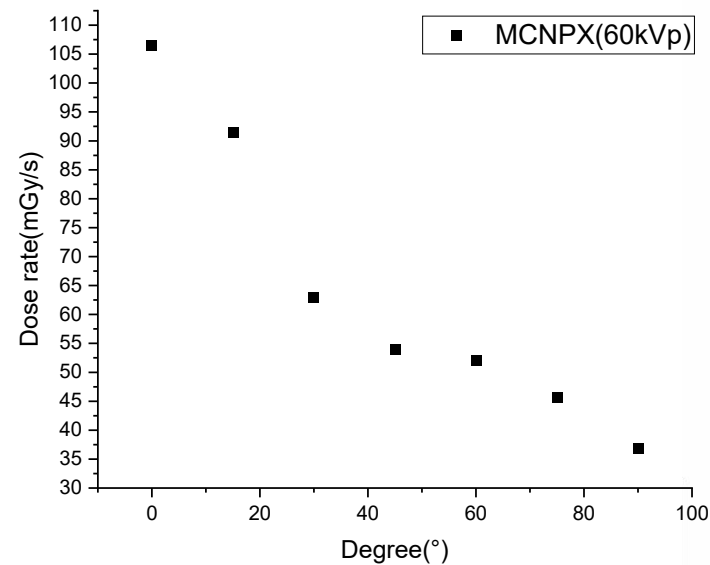
4.3 X-ray based SILPS Calibration

4.3.6 Angular Dependence

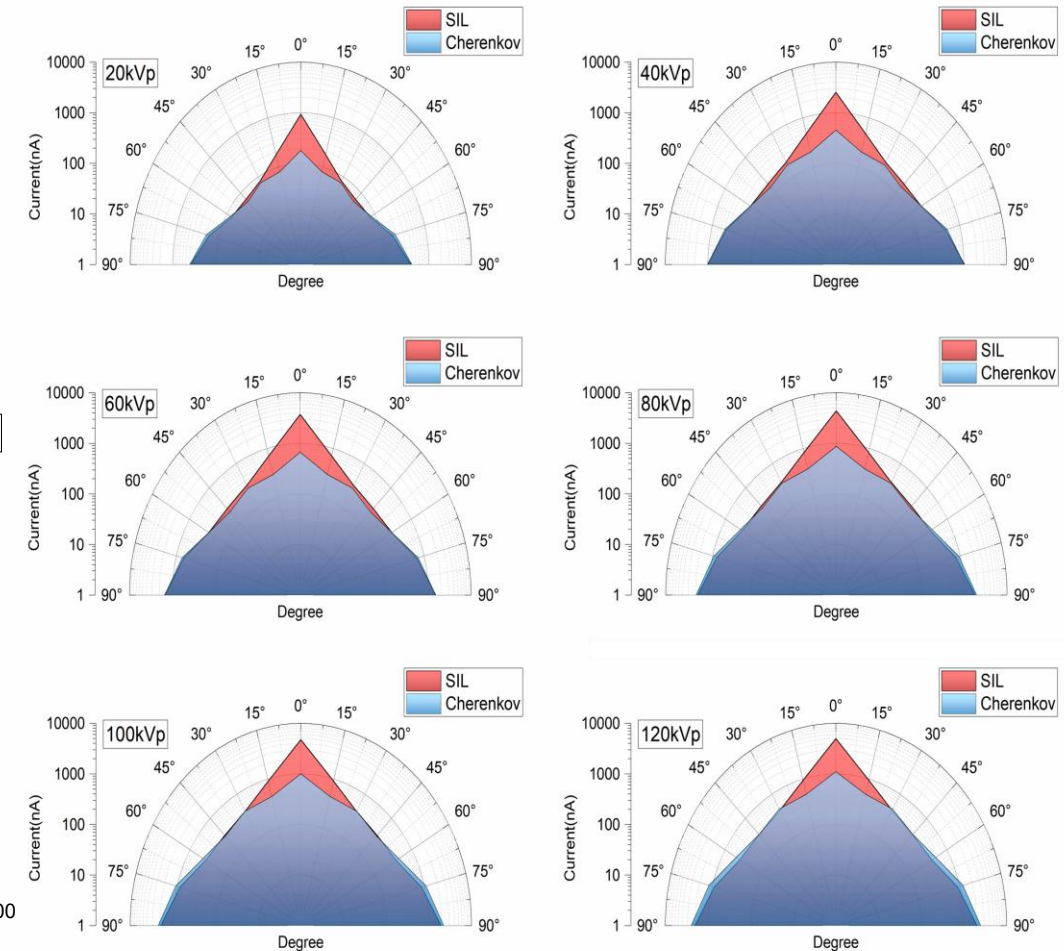
- Angular Dependence of SILPS Dose Rate Measurement for X-rays.
- (Will be updated) SILPS results are being compared with Monte Carlo simulations to verify accurate dose rate measurements under different X-ray incident angles



< Angular setup >



< MCNPX simulations >



< Angular dependency of SILPS >

5. Conclusion

- This research presents a 3D-printing based system for localized skin dosimetry aligned with the ICRP definition (nominal depth $\sim 70\ \mu\text{m}$ and averaging over $1\ \text{cm}^2$).
- A DLP-printable plastic scintillator was developed and optimized, including formulations with longer emission wavelengths to improve optical readout performance in SiPMs.
- A Skin Imitating Layer (SIL) consisting of a $50\ \mu\text{m}$ epidermis layer (EL) and a $50\ \mu\text{m}$ basal layer (BL) was designed to physically realize the sensitive skin depth region.
- A continuous multi resin DLP process enabled fully bonded EL/BL fabrication, and the layer thickness, uniformity, reproducibility, and light induced stability were experimentally validated.
- The tissue equivalence of the SIL was assessed using composition-based metrics and Monte Carlo simulations, and compared with ICRP reference materials.
- A SIL Phantom System (SILPS) with current mode readout was implemented, and a practical procedure for radioluminescence separation was established.
- X-ray experiments demonstrated system feasibility through calibration and validation, including dose linearity, current to dose rate conversion, method comparison, and angular dependence evaluation.

Thank you